

KLM · SSA-3

PATHOGEN SURVEILLANCE

OUTBREAK-TRACING ENGINE

MODULES

Overview

Genome Indexing

Relatedness

Mutations

Transmission Network

Clusters

Spread Simulator

Budget Optimizer

Alerts

Samples

SEED 20260921
126 GENOMES · 503 BP
A · T · G · C

SYSTEM OVERVIEW

OVERVIEW

Q Search sample, cluster or loc

ENGINE ONLINE · 503 EDGES

SYNTHETIC DATA — PROTOTYPE

1

SYSTEM PIPELINE · LIVE STATE

OVERVIEW

Every figure on this page is computed in-browser from the synthetic genome set using the same algorithms exposed on the module pages. Changing the relatedness threshold or scoring method recomputes the entire chain.

t = 0.74METHOD · KMERk = 5OPEN NETWORK

Genome Data · Index · Compare · Detect · Graph · MST · Cluster · Bridges · Simulate · Optimize · Alert

MONITORED SAMPLES

+1,248

126 genomes in the interactive engine

OUTBREAK CLUSTERS

+10

10 unlinked singletons at t = 0.74

DISTINCT MUTATIONS

+656

vs reference genome · top M-001 @ 154

CRITICAL LINKS

+0

Tarjan bridges in the transmission graph

NETWORK EDGES

+503

density 6.39% · avg degree 8.0

MST TOTAL WEIGHT

+15.827

106 backbone edges (Kruskal)

SEQUENCING BUDGET

-

run the greedy optimizer to allocate

RECOMMENDED SAMPLES

+15

heuristic estimate until optimizer runs

TRANSMISSION NETWORK

126 nodes · 503 edges · bridges dashed red

FULL VIEW

ACTIVE ALERTS

9

Dense outbreak cluster CL-01HIGH

12 samples at density 83% (avg similarity 0.799) centred on Hyderabad.

ALR-001 DENSE_CLUSTER 25 Sept., 08:43 am

STATUS · OPENCL-01

Dense outbreak cluster CL-02HIGH

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ALGORITHM 01 · K-MER INDEX

GENOME INDEXING

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SYNTHETIC DATA — PROTOTYPE

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ALGORITHM 01 · INVERTED INDEX

GENOME INDEXING

Every genome is decomposed into overlapping k-mers and stored in an inverted index. Changing k discards the index and rebuilds it from the raw sequences — the statistics below are read straight off the rebuilt structure.

k = 3k = 5k = 7

INDEXED GENOMES

126

320 bp each · alphabet A · T · G · C

TOTAL K-MERS

39,816

sliding-window instances across the cohort

UNIQUE K-MERS

1,006

98.2% of the 4^k = 1,024 space

AVG POSTING LENGTH

33.47

33,666 total postings in the index

K-MER LOOKUP

findSamplesByKmer(index, kmer) · O(1) average

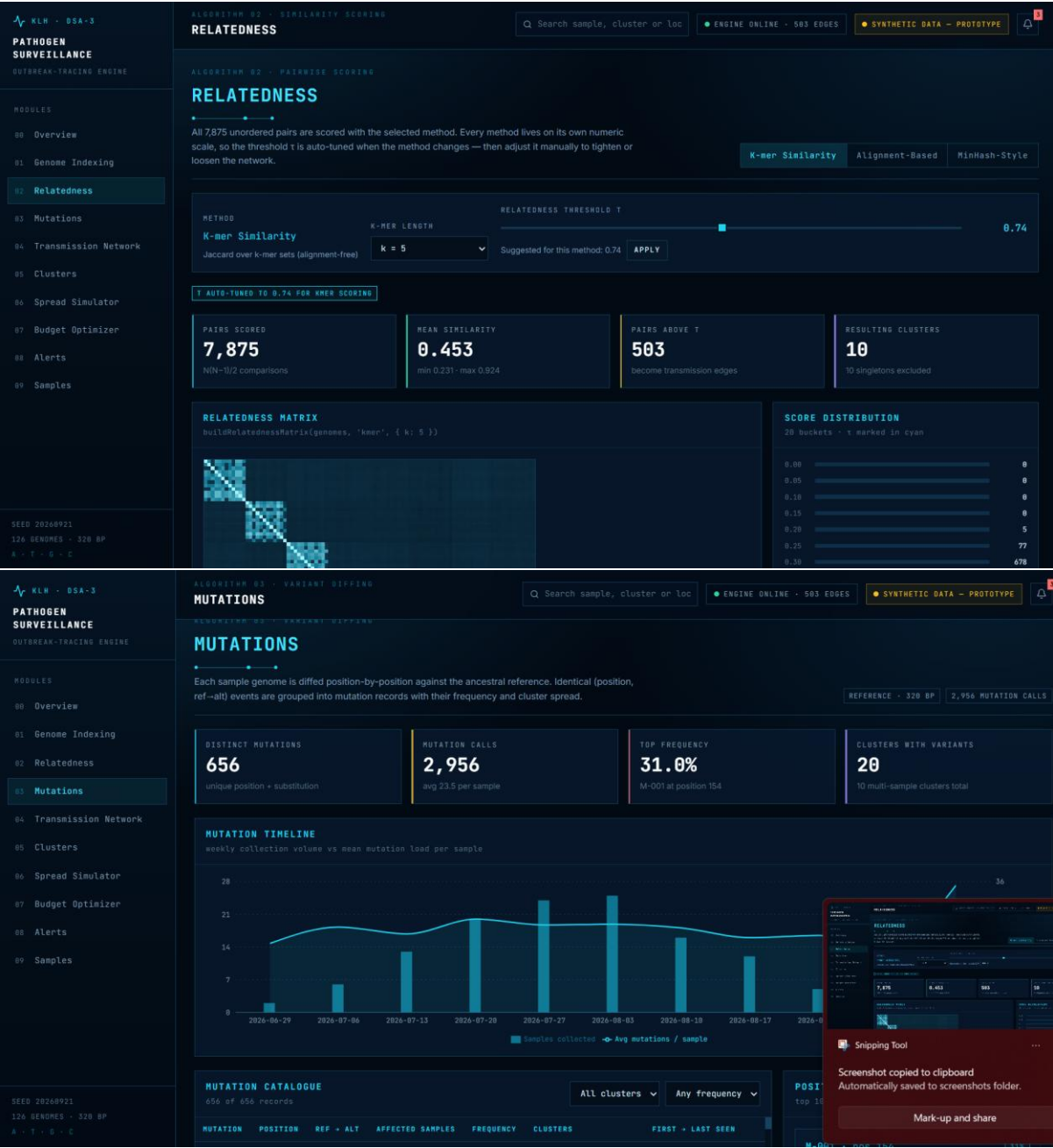
Q Type a 5-mer, e.g. ATGCA

Enter a 5-base string to retrieve its posting list from the index.

MOST FREQUENT K-MERS

ranked by posting-list length

TATTC	126
CAGTT	125
GGCCA	125
AACCC	124
ATTCT	124
CTGGT	124
TACTA	124
GCCAT	123



ALGORITHM 03 · VARIANT DIFFING

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SYNTHETIC DATA — PROTOTYPE

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MUTATIONS

ALGORITHM 03 · VARIANT DIFFING

Each sample genome is diffed position-by-position against the ancestral reference. Identical (position, ref→alt) events are grouped into mutation records with their frequency and cluster spread.

REFERENCE · 320 BP · 2,956 MUTATION CALLS

DISTINCT MUTATIONS

656

unique position · substitution

MUTATION CALLS

2,956

avg 23.5 per sample

TOP FREQUENCY

31.0%

M-001 at position 154

CLUSTERS WITH VARIANTS

20

10 multi-sample clusters total

MUTATION TIMELINE

weekly collection volume vs mean mutation load per sample

MUTATION CATALOGUE

656 of 656 records

All clusters

Any frequency

POSITION

TOP 10

MUTATION	POSITION	REF · ALT	AFFECTED SAMPLES	FREQUENCY	CLUSTERS	FIRST · LAST SEEN
...						

SEED 20260921

126 GENOMES · 320 BP

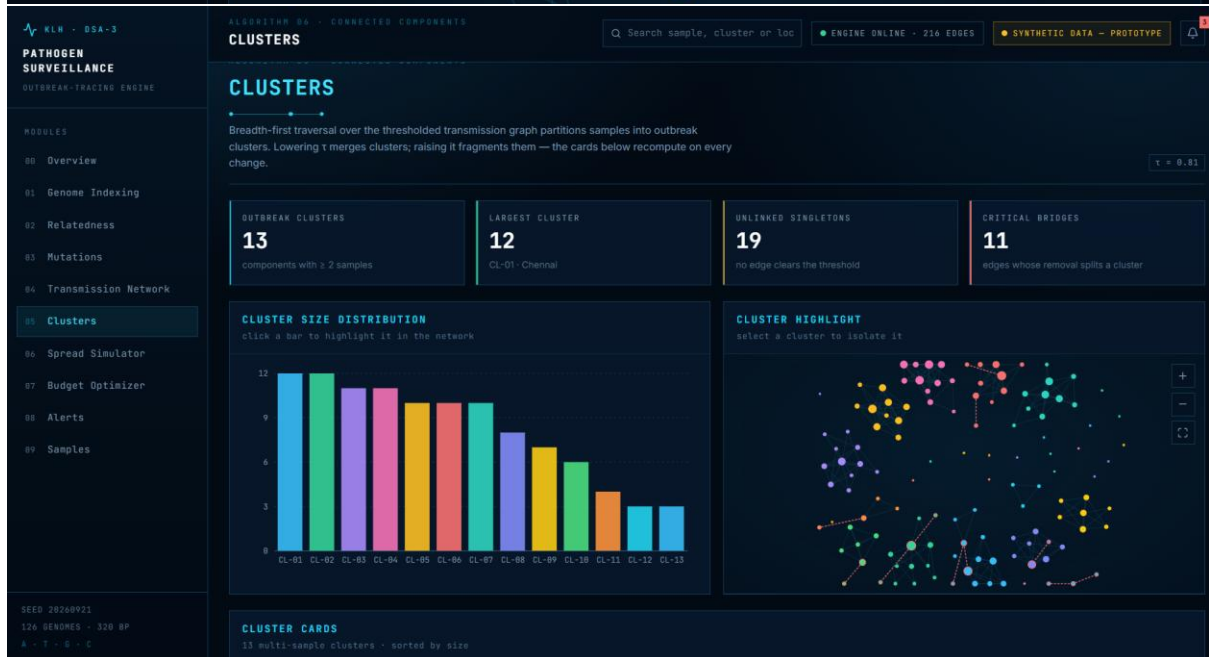
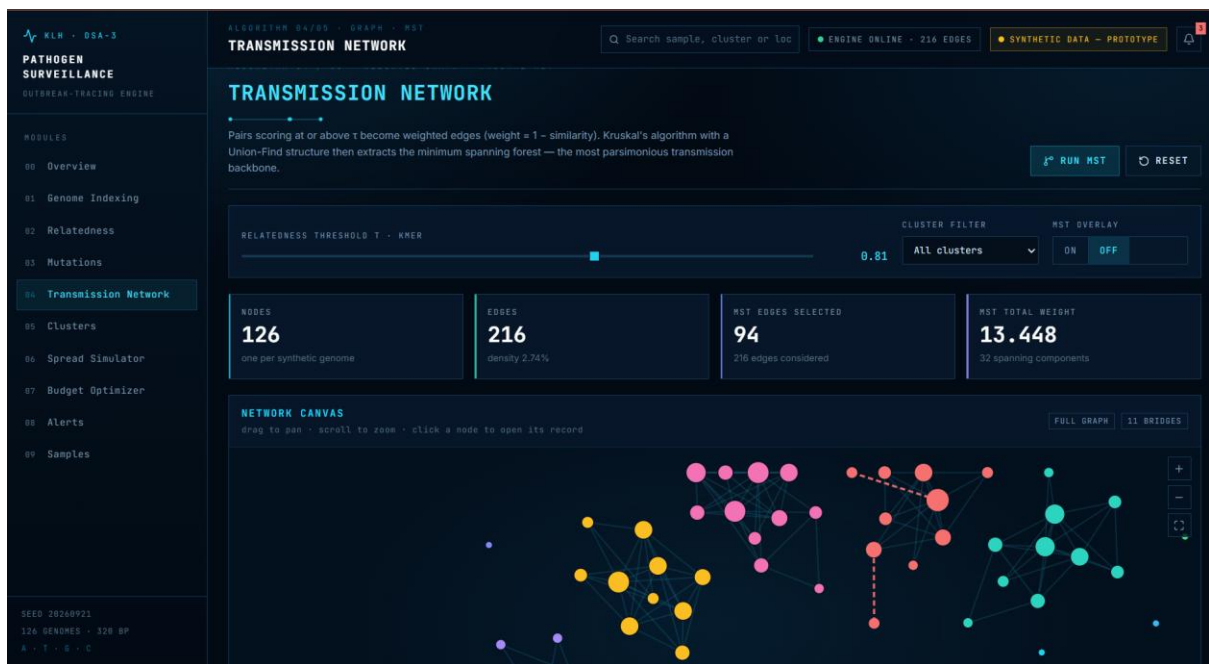
A · T · G · C

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Mark-up and share



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00 Overview

01 Genome Indexing

02 Relatedness

03 Mutations

04 Transmission Network

05 Clusters

06 Spread Simulator

07 Budget Optimizer

08 Alerts

09 Samples

SEED 20240921

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ALGORITHM: DT/DS · TARJAN · ISOLATION

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SPREAD SIMULATOR

Tarjan's DFS flags the edges holding the network together, then the isolation simulator removes a sample and recomputes connected components from scratch. Nothing here is hardcoded — both panels read the live graph.

DETECT CRITICAL LINKS

RESET

CRITICAL LINKS

11

detection runs: 0

COMPONENTS

32

largest holds 12 samples

LARGEST COMPONENT SHARE

9.5%

avg degree 3.43

ISOLATED SAMPLES

19

at t = 0.81

CRITICAL LINK TABLE

run detection to list the bridge edges

TABLE

PRESS "DETECT CRITICAL LINKS" TO RUN TARJAN ON THE CURRENT GRAPH

SAMPLE TO ISOLATE

Select a sample...

SIMULATE ISOLATION

HIGHEST-IMPACT ISOLATION CANDIDATES

each candidate is actually simulated · ranked by component fragmentation

SAMPLE	DEGREE	LINKS BROKEN	NEW COMPONENTS	FRAGMENTATION
S-044	9	9	3	-0.08%
S-035	8	8	1	-0.08%

ALGORITHM: GP · GREEDY SET COVER

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SYNTHETIC DATA - PROTOTYPE

1

BUDGET OPTIMIZER

Each candidate sample carries a sequencing cost and a coverage set drawn from live clusters, mutations, bridges and network neighbourhoods. The optimizer repeatedly takes the best marginal-coverage-per-rupee candidate that still fits the remaining budget.

RUN GREEDY OPTIMIZATION

RESET

AVAILABLE BUDGET

UNIVERSE

825 elements

126 affordable candidates

SAMPLES SELECTED

—

greedy selection under budget

TOTAL COST

—

budget ₹60,000

COVERAGE

—

run the optimizer

CRITICAL LINKS COVERED

—

11 bridges in the universe

ITERATION TRACE

marginal coverage + cost at every step

NO RUN YET — PRESS "RUN GREEDY OPTIMIZATION"

COVERAGE BREAKDOWN

Clusters covered	—
Mutations covered	—
Network nodes covered	—
Critical links covered	—